

LOKESH PARASURAMAN

Software Developer

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PROFILE SUMMARY

Motivated Computer Science and Engineering graduate with strong foundations in **Java, Data Structures & Algorithms (DSA)**, and web application development. Passionate about designing optimized backend systems, writing clean code, and solving complex algorithmic challenges. Eager to apply problem-solving technical skills and a **quick-learning** mindset to contribute effectively to software engineering teams.

TECHNICAL SKILLS

- **Programming Languages:** Java (Core & Advanced), JavaScript, TypeScript
- **Core Competencies:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Problem Solving
- **Backend & System Design:** Node.js, RESTful APIs, Role-Based Access Control (RBAC), System Architecture
- **Frontend Engineering:** React.js, HTML5, CSS3, Tailwind CSS
- **Databases & Storage:** PostgreSQL, MySQL, Database Management Systems (DBMS), Query Optimization
- **Generative AI & Cloud:** OpenAI API Integration, Docker, Cloud Deployment, GitHub Actions

TECHNICAL PROJECTS

Fly-Fast – AI-Assisted Flight Booking & Management Platform

Technologies: Node.js, React.js, PostgreSQL, Docker, GitHub Actions, OpenAI API

- Engineered an end-to-end flight booking platform covering search, booking, cancellation, and automated refund workflows.
- Optimized application workflows using algorithmic data structuring to reduce multi-step user search friction by **approximately 60%**, incorporating a conversational **AI assistant via the OpenAI API**.
- Designed and implemented database-layer **concurrent seat-availability locks** to **eliminate double-booking** race conditions under simultaneous heavy transactional loads.
- Developed a resilient payment processing module supporting UPI and card transactions equipped with real-time state tracking, automatic failed-payment retries, and cancellation policy enforcement.
- Architected the complete booking lifecycle management system incorporating automated PNR generation, transactional seat reservation states, and an audit-friendly historical ledger.
- Containerized the layout using Docker and established a fully automated CI/CD pipeline via GitHub Actions for production cloud deployment.

Cryptographic Chain of Custody & Digital Evidence Tracking System

Technologies: Node.js, MySQL, SHA-256 Cryptography, Role-Based Access Control (RBAC), Git

- Conceptualized and engineered a secure digital evidence tracking system implementing SHA-256 cryptographic hashing to ensure absolute data integrity and **tamper-proof records**.
- Achieved the architectural properties of blockchain immutability and linear compliance tracking within a relational database design, mitigating processing overhead.
- Designed and executed a strict Role-Based Access Control (RBAC) mechanism encompassing Investigators, Administrators, and Auditors to guarantee adherence to the principle of least privilege.
- Built an independent cryptographic verification module enabling system stakeholders to programmatically validate any historical evidence record against its original hash to identify anomalies.
- Optimized underlying database queries and schema indexes in MySQL to handle extensive historical audit logs **without degradation** in retrieval times.

EDUCATION

Bachelor of Engineering (B.E.) – Computer Science & Engineering

Coimbatore Institute of Engineering and Technology (2021 – 2025) | CGPA: 7.58 / 10

- **Relevant Coursework:** Data Structures & Algorithms (DSA), Database Management Systems (DBMS), Object-Oriented Software Engineering, Operating Systems.

TECHNICAL ACHIEVEMENTS & RECOGNITIONS

- **First Prize Winner:** Secured top honors for the final year technical project presentation evaluation among competing engineering cohorts.
- **Good Quality Code Award:** Officially recognized for outstanding architectural formatting, clean code practices, and optimal implementation efficiency during the Inter-Collegiate Hackathon hosted at SNS College of Engineering.

